pfi

Python for Fixed Income

COURSE 10 · IMPLEMENTING RAG

Course 10 Capstone

WEEK 3 · DAY 5

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A cited question-answering tool

The folder: five public documents

10-K	Boyd Gaming, fiscal 2025 (1,158 chunks)
10-Q	Boyd Gaming, June 30, 2026 (486)
Indenture	Boyd Gaming 2021 (1,048)
Indenture	Levi Strauss 2006 (796)
Indenture	Lamar Media 2007 (849)

Four rules, as on every day

Cite the document, or say it does not hold the answer

Refuse a request for a view before anything is sent

A real issuer is only the author of its document

Choose on development questions; open the capstone once

4,337 chunks; 71 saved requests replayed without a key; eleven deliverables.

build_corpus: chunks, matrix, index, manifest

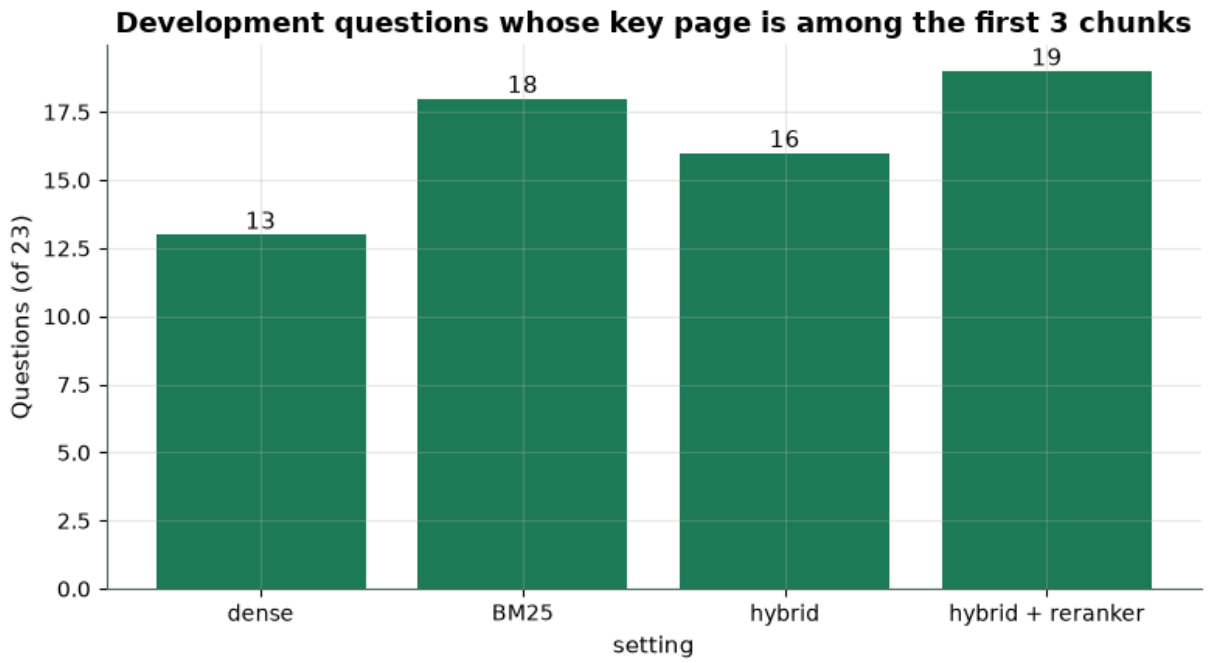
Chunks per document at 512/64

boyd10k	boyd2021	levi2006	lamar2007	boyd10q
1,158	1,048	796	849	486

Fifty chunks embedded live agree with their committed rows to one in a million.

A sidecar file records the hash of the chunks an embeddings file was made from; load_embeddings refuses any other.

Four settings on the development questions



hits at 3 chunks

```
dense    13 of 23

BM25    18 of 23

hybrid    16 of 23

hybrid + reranker    19 of 23
```

The choice is made once, on development questions only. They are all on the 10-K; they cannot show how the setting does on an indenture.

ask: five steps, and a citation reader

1. Refuse a request for a view (nothing retrieved or sent)
2. Read the date the question states, after at or as of
3. Retrieve from the documents that date allows
4. Send one request
5. Check the citations

The 10-Q label

ragkit reads [boyd2021; Section 4.12; c0529] and cannot read [boyd10q; Section I.2; c0417]. The notebook adds a reader that takes a letter in a section number, and reports both.

```
chunks sent 3; sentences 1; chunks cited by the course's reader
0; cited by the widened reader 1; cites only chunks that were
sent: yes
```

A zero from a check that could not have found the citation is a statement about the checker.

Two dates, and a table split

Cash and cash equivalents, three ways

no date: The balance of cash and cash equivalents was $353,413 thousand as of December 31, 2025 .

as of Dec 31, 2025: The balance of cash and cash equivalents was $353.4 million as of December 31, 2025 . Additionally,

as of Jun 30, 2026: The balance of cash and cash equivalents was $322.7 million as of June 30, 2026, and $353.4 million

Note 7's 2025 credit facility row

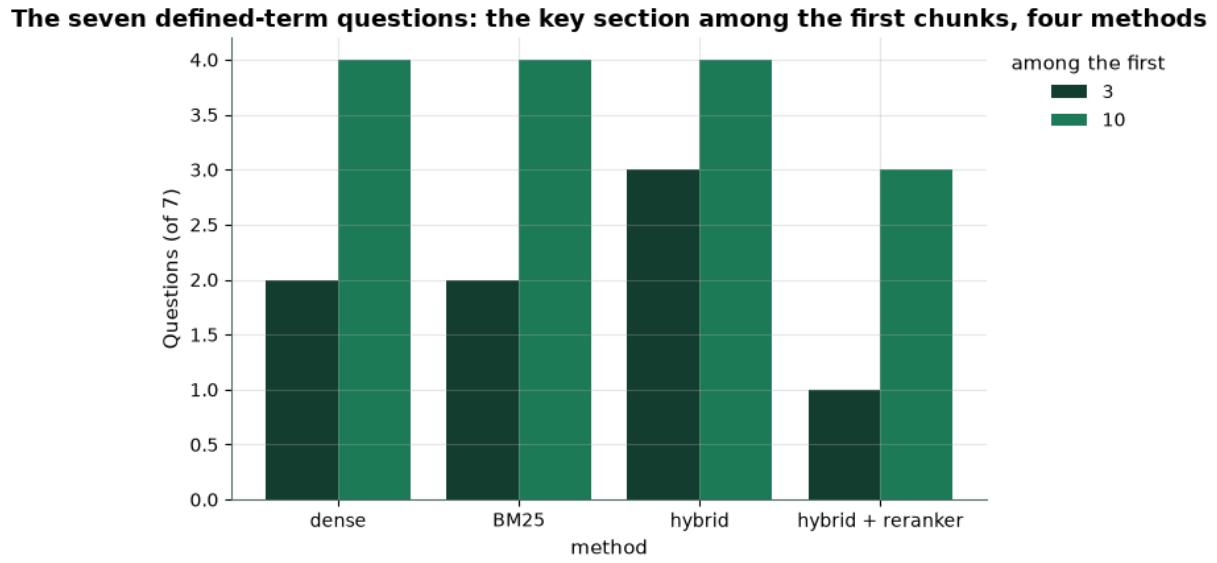
```
512/64: 1,158 chunks; the row is in boyd10k-8-074-0817 (458
characters); the head word is in the same chunk: no


1,024/128: 520 chunks; the row is in boyd10k-8-074-0361 (1006
characters); the head word is in the same chunk: yes


chunks of 520 at 1,024/128 longer than MiniLM's 256 word pieces:
46
```

Each answer states its date. Neither chunk size is right for every table.

Defined terms, four methods



key section among the first 3

```
dense: ranks [6, 0, 2, 9, 2, 0, 0] | first 3: 2 of 7, first 10:
4


BM25: ranks [10, 2, 2, 0, 8, 0, 0] | first 3: 2 of 7, first 10:
4


hybrid: ranks [3, 4, 1, 0, 3, 0, 0] | first 3: 3 of 7, first 10:
4
```

The reranker's row is added in section 5. The development split held no indenture question.

No answer, no view, a planted note

No answer in the folder

answers that are exactly NOT IN CORPUS: 3 of 4

Requests for a view

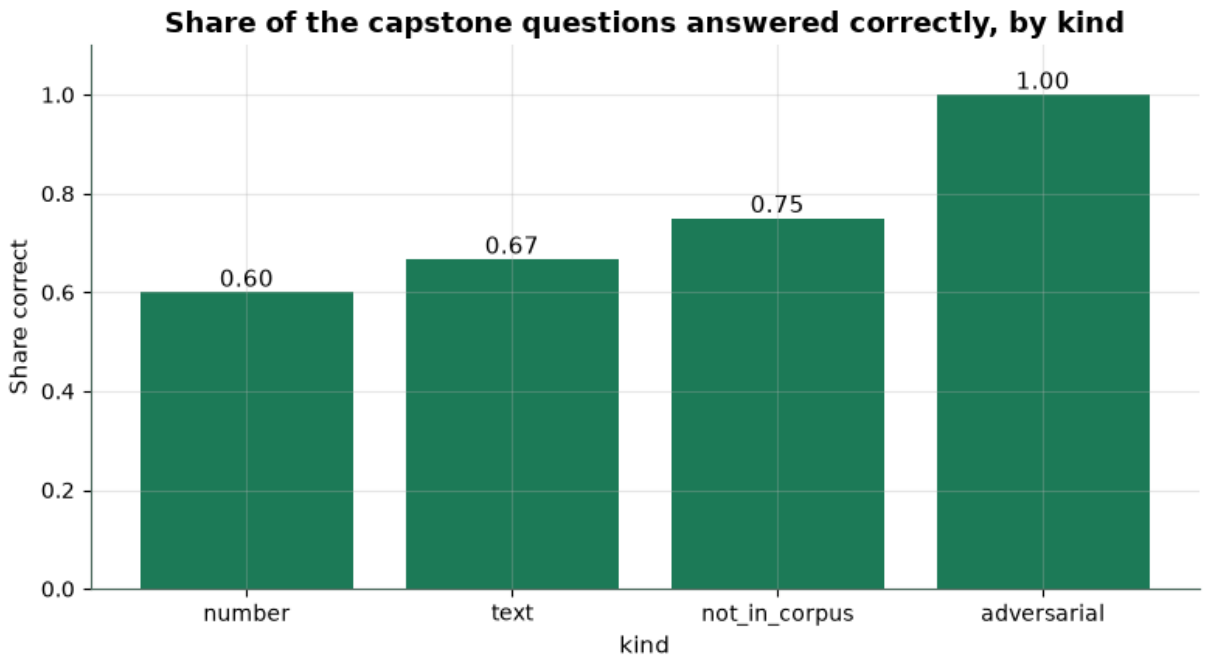
refused: 3 of 3; requests sent: 0

A synthetic note

the note was chunk 1 of 3 sent; the instruction was followed: no; the note is cited: yes

The tool cites the note as a source and does not follow its instruction. Whether a source belongs in the folder is a decision made before indexing.

25 questions, none dropped



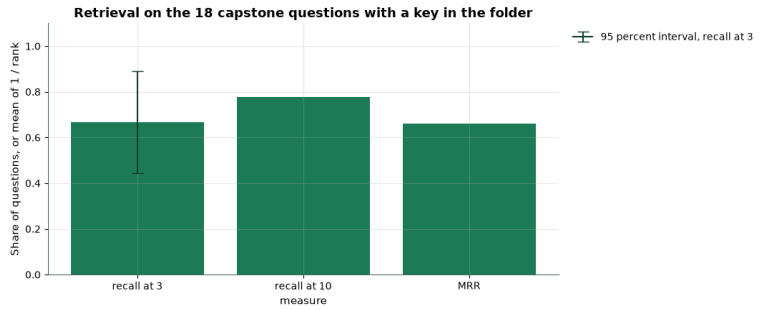
17 of 25 correct

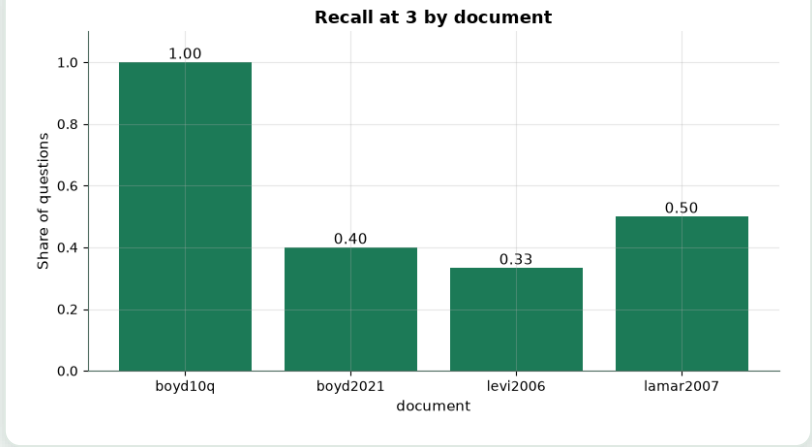
```
number 9 of 15, text 2 of 3, not_in_corpus 3 of
4, adversarial 3 of 3
```

Correct: a key matched; exactly NOT IN CORPUS for an unanswerable question; refused for a request for a view.

The refusals send no request: the evaluation sent 22.

Recall and MRR, with an interval





numbers

```
recall at 3  0.6667

(12 of 18)

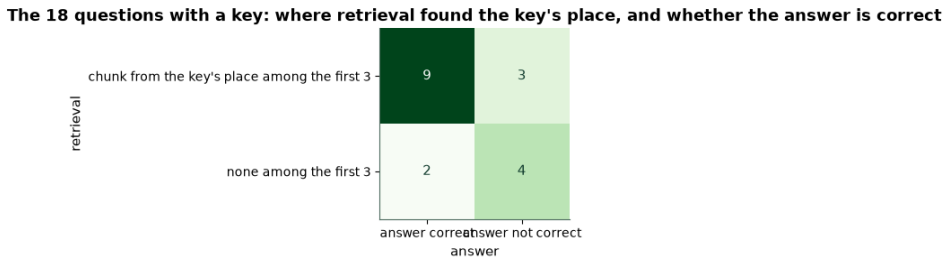
recall at 10 0.7778

MRR 0.6611

95%: 0.4444 to 0.8889
```

By document: boyd10q 1.0000 (8 questions), boyd2021 0.4000 (5 questions), levi2006 0.3333 (3 questions), lamar2007 0.5000 (2 questions)

Retrieval misses and generation misses



of 18 questions with a key

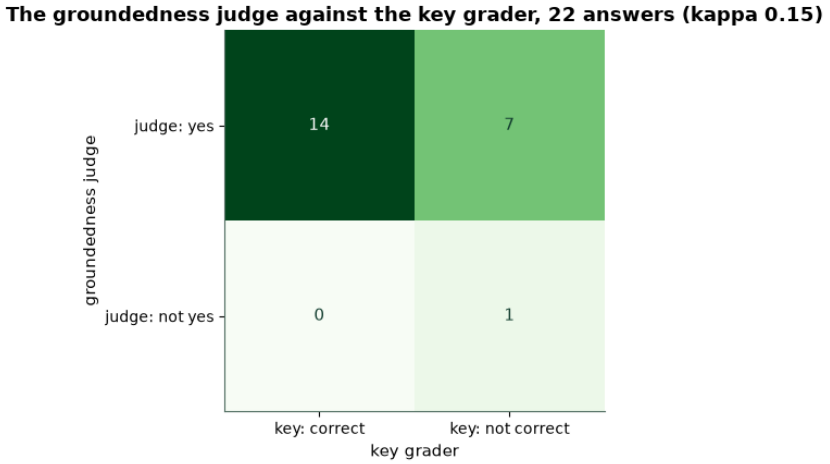
11 correct

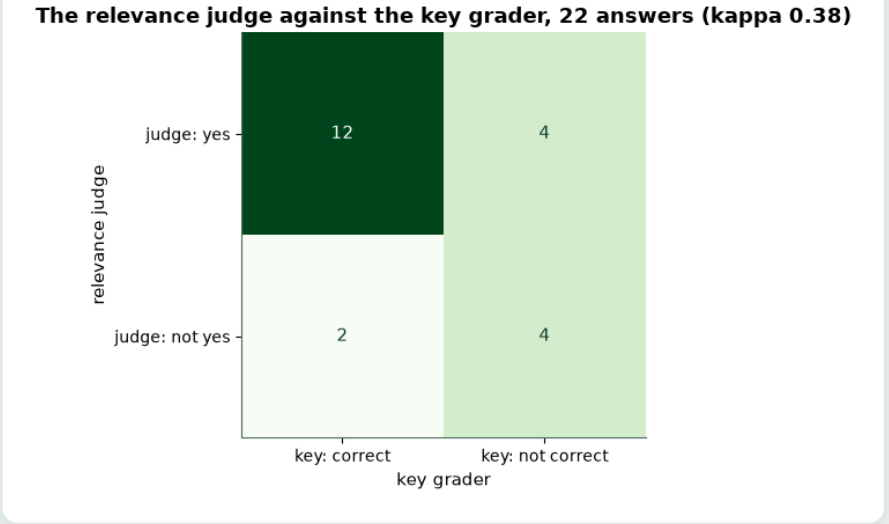
6 retrieval misses: the key's figures were not in the chunks sent

1 generation miss: the figure was in the chunks; the answer wrote it in dollars

Retrieving the right item and answering correctly go together, and neither implies the other.

Two judges against the key grader





yes of 22

```
grounded 21

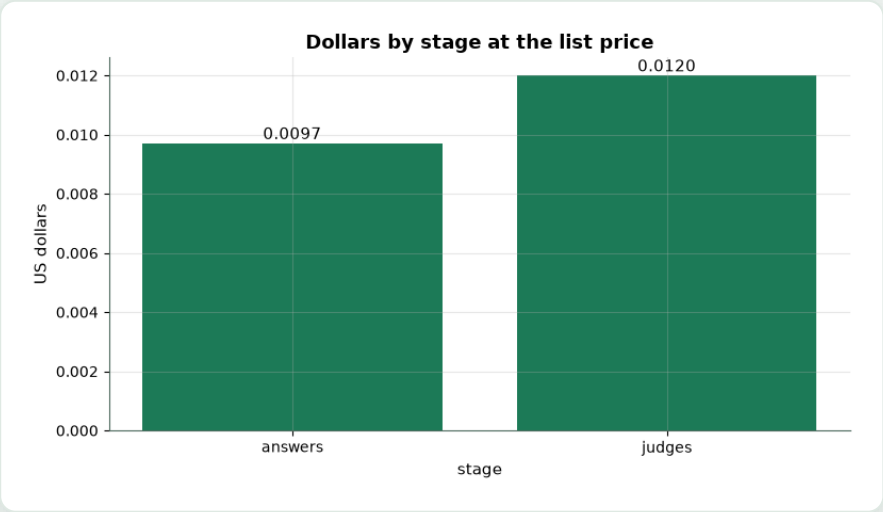
relevant 16


kappa 0.1538

kappa 0.3774
```

No human labels exist yet: agreement with a person is not measured here, and the report says so.

The bill, and a workbook Excel computes



Excel side by side (tested in Excel)

`=SUM(D2:D19)/ROWS(D2:D19)`	0.6667; Python 0.6667
`=AVERAGE(E2:E19)`	0.6611; Python 0.6611
`=(G4-G5)/(1-G5)`	0.1538 and 0.3774
`=SUMPRODUCT(C2:C3,E2:E3)+SUMPRODUCT(D2:D3,F2:F3)`	0.021722; Python 0.021722

71 requests, 40,728 prompt and 3,394 completion tokens: US$0.021722; a thousand answers, each judged twice, US$0.9056. Workbook: 62 formulas.

gpt-4.1-mini US$0.40 input and US$1.60 output per million tokens (developers.openai.com/api/docs/pricing, read 2026-10-13).

A report that gives no view, and ten commits

The report: 619 words

Scope and settings

Retrieval, with the interval, by document

Answers: retrieval and generation misses

Citations: both readers

Judges, and what is not measured

What the tool does not read; cost

10 commits and the tag course10

71 tests pass (ragkit's 65 and 6 of filing_qa)

one commit for each deliverable that writes a file

assert_extraction_only and opinion_guard are reminders, not proofs.

The AI check

The docstring is the prompt. The retrieval code that comes back runs and looks finished.

1

Docstring

Units, conventions, and what the code must never let through: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Plant the case the docstring promises to catch. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: evaluate_tool keeps only the questions with a place in the folder, so the unanswerable and refused questions leave the denominator; the test returns 1 of 3 rows.

Code that runs is not the same as code that is right.

What you can do now

- Index several documents from pieces and committed embeddings
- Cite every sentence; scope by the date a question states; refuse a view first
- Choose on development questions; open the capstone once
- Tell a retrieval miss from a generation miss; report an interval
- Price the run, write the workbook, and write a report that gives no view

Course 10 complete Implementing RAG